ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY

MODULE 2,TERM 2

UNIT: NETWORK SETUP

UNIT CODE: 0612 351 03A

TIME 2 HOURS

TOTAL 30 MARKS

Materials and tools

- ✓ 2 m of Cat5e/Cat6 UTP cable (3 pieces)
- ✓ 6 RJ45 connectors (with boots, if available)
- ✓ Wire stripper and cutter
- ✓ Crimping tool
- ✓ Network cable tester (with remote unit)
- ✓ Digital multimeter
- ✓ Measuring tape
- ✓ Pen and the answer sheet below
- ✓ Rim paper /foolscap for sketching

Instructions to candidates

- ✓ Attempt all tasks in order.
- ✓ Record all observations on the answer sheet.
- ✓ Wear safety glasses when cutting and crimping.
- ✓ Call the examiner to verify Tasks 2, 4 and 6 before moving on.

PRACTICAL TASKS

Task 1: Cable preparation using the wire stripper and cutter

1. Cut one piece of cable to exactly 1 m using the cutter.
2. Strip about 2.5 cm of outer jacket from both ends without nicking the inner conductors.
3. Untwist the pairs and straighten the wires.

Task 2: Crimping an RJ45 straight-through cable

1. Arrange the wires in the **T568B** order at both ends:
   white/orange, orange, white/green, blue, white/blue, green, white/brown, brown
2. Trim the wires evenly, insert them fully into the RJ45 connector (clip facing away), and crimp.
3. Repeat at the other end.

Task 3: Continuity testing with a multimeter

1. Set the multimeter to **continuity (beep)** mode, or to the lowest resistance range.
2. Touch the probes to the same pin position at each end of the cable (pin 1 to pin 1, and so on).
3. Record the result for each pin.
4. Test one pin against a different pin (e.g. pin 1 to pin 2) to check for shorts.

Observation table

Pin	Beep / Resistance (Ω)	Continuity (Yes/No)
1		
2		

3		
4		
5		
6		
7		
8		

Task 4: Wire mapping with a cable tester

1. Plug each end of your crimped cable into the main and remote units of the tester.
2. Switch on the tester and observe the LED sequence.
3. Record which LEDs light and in what order.
4. Sketch the wire map showing which pin at end A connects to which pin at end B.
5. State whether the cable passes or fails, and name the cable type.

Task 5: Cable length testing

1. Measure the cable physically with the measuring tape.
2. If your tester supports length measurement, record its reading. If not, state how a TDR-type tester measures length.
3. Compare your measured and tester values, and state the maximum permitted Ethernet segment length.

Measurement	**Value**

Tape-measured length	
Tester-measured length (if available)	
Difference	
Maximum permitted length (standard)	

Task 6: Fault detection and rectification

The examiner will give you a **faulty cable** with one or more of these faults: open wire, short, reversed pair, split pair, or miswire.

1. Test the cable with the cable tester and multimeter.
2. Identify and record the fault type and the affected pins.
3. Rectify the fault by re-terminating (cut, strip, re-crimp).
4. Retest to confirm the cable now passes.

The trainer is introducing oral questions shortly.